

PixelCatsFork

Project navigation

Use this quick map to find the main entry points in the repo:

- `PixelGame.sln` — open this solution to load all primary projects.
- `PixelBoardDisplay/` — shared display/input abstractions and board integration code.
- `ConsoleTest/` — console-hosted game runner and game implementations.
- `ConsoleTest.Tests/` — unit tests for console/game and API client behavior.
- `HerdingCats/` — standalone game project.
- `PixelCatsClient/` and `Client/` — client applications.
- `shared/latest_score.json` — shared score output consumed by parts of the solution.

Quick start by goal

- **Run core console game loop:** start in `ConsoleTest/Program.cs`.
- **Work on board rendering or Arduino integration:** start in `PixelBoardDisplay/`.
- **Run or add tests:** start in `ConsoleTest.Tests/`.

Prerequisites

- .NET SDK 9.x
- Visual Studio 2022 (optional)

Build / run / test

From the repo root:

```
dotnet build PixelGame.sln
dotnet run --project ConsoleTest/ConsoleTest.csproj
dotnet test
```

Configuration

`ConsoleTest` loads configuration from `appsettings.json` (optional) and environment variables.

- `UseEmulator` (bool) — defaults to `true`
- `Leaderboard:BaseUrl` (string) — defaults to `http://127.0.0.1:3000`
 - Environment variable form: `Leaderboard__BaseUrl`
- HMAC secrets:
 - `LEADERBOARD_HMAC_SNAKE`
 - `LEADERBOARD_HMAC_TETRIS`
 - `LEADERBOARD_HMAC_EDU`

Score export

`ConsoleTest` writes to:

```
shared/latest_score.json
```

This path is relative to the repository root when possible.

Adding a new game

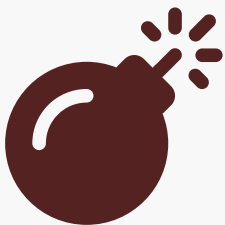
To add a new game in `ConsoleTest`:

1. Implement `ConsoleTest/Games/IGame.cs`.
2. Register the game in `ConsoleTest/Program.cs` inside the `games` dictionary.

Class diagrams

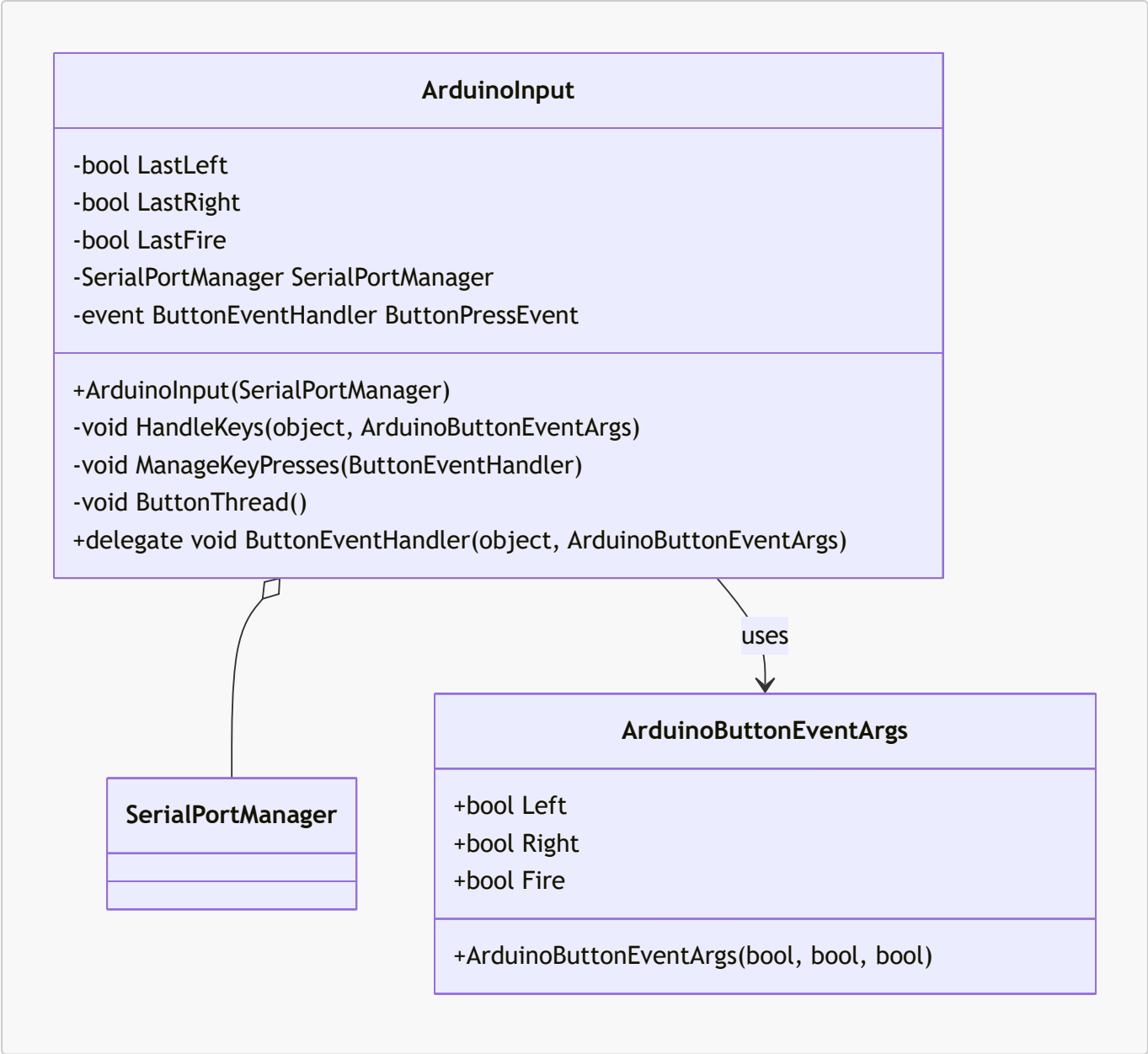
Below are class diagrams, in Mermaid format, for the major classes and interfaces in the PixelBoard project.

ArduinoDisplay

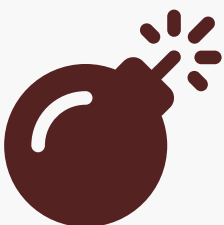


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ArduinoInput



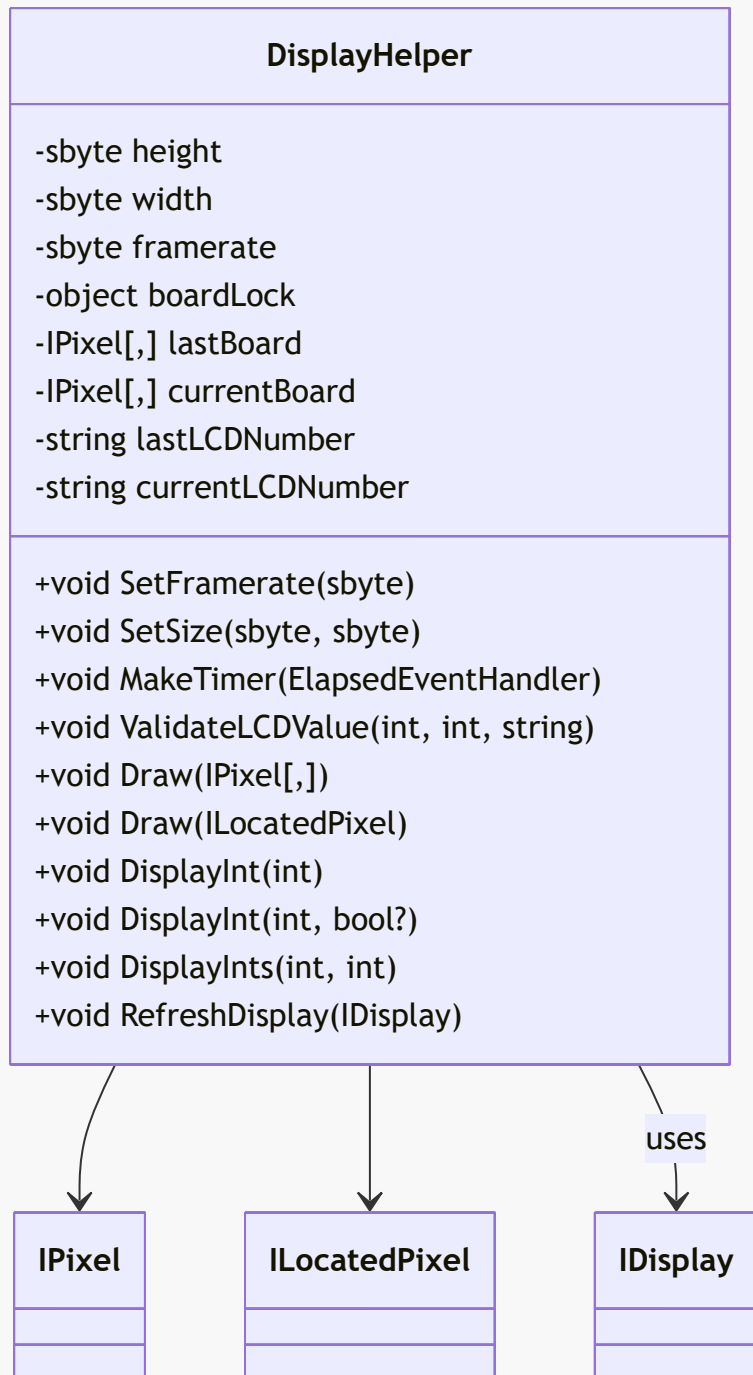
ConsoleDisplay



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DisplayHelper



IArduinoInput



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IDisplay



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ILocatedPixel



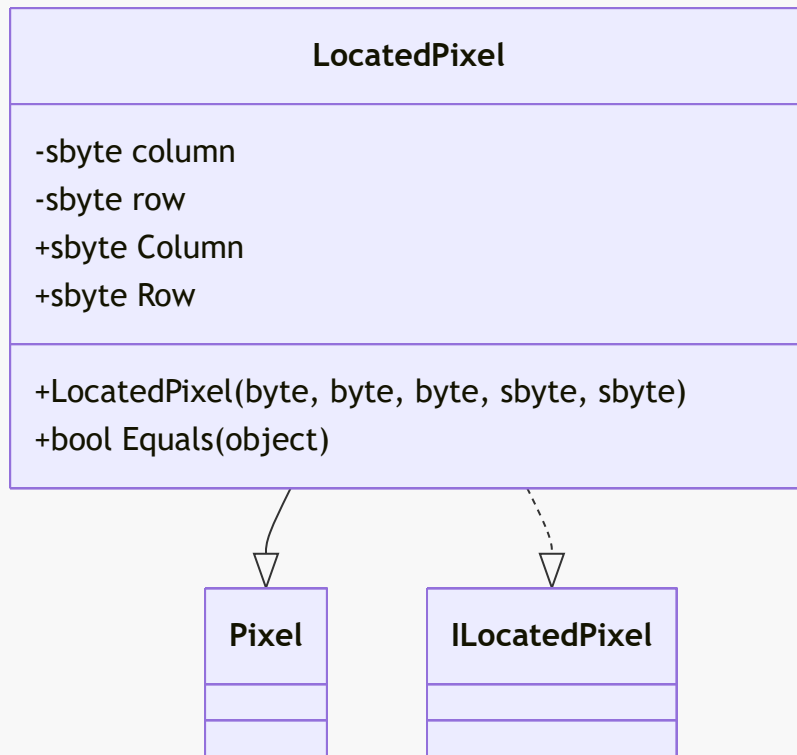
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IPixel

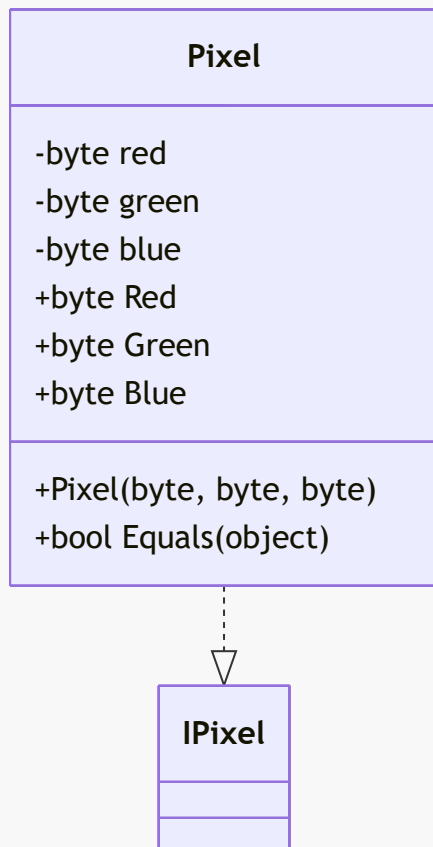


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LocatedPixel



Pixel



SerialPortManager

